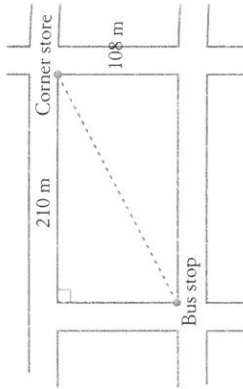


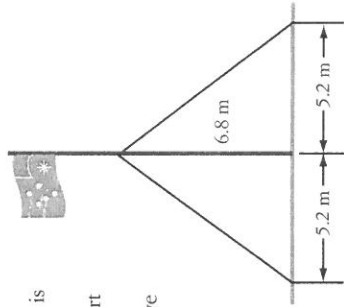
Exercise 9.3 Applications of triangles

Problem solving

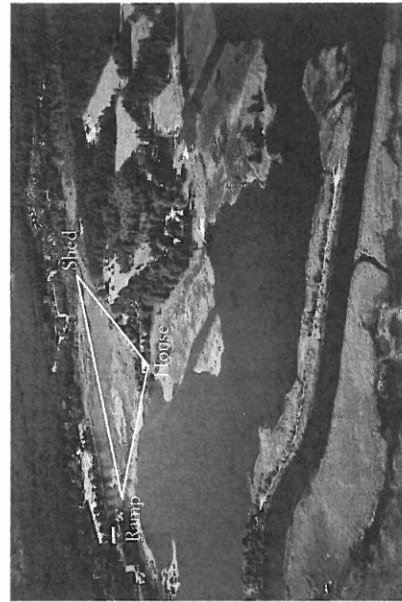
- 1 A bus stop and a corner store are at diagonally opposite corners of a suburban block as shown. Tammy gets off the bus and takes a short cut to the store on her way home from school. Find:
a the distance she travels by taking the short cut
b the distance she saves by not going the long way to the corner store.



- 2 A flagpole is supported by two guy-wires that are attached to the pole 6.8 m above the ground. Each wire is attached to the ground 5.2 m from the base of the flagpole. Find the total length of the wire used to support the flagpole if an additional 30 cm is required for each guy-wire to attach them to the pole and the ground. Give your answer correct to the nearest centimetre.



- 3 Jose is a keen fisherman who lives on a lake. As shown in the photograph below, 670 m directly west of his house is the boat ramp he uses when he launches his boat into the water. 820 m directly north of the boat ramp is a shed that he uses to store fishing gear and carry out minor repairs to his boat. How far is the shed from Jose's house?

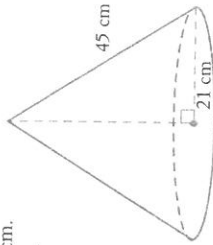


See Example 15

Extra questions
Exercise 9.3
MAT09MGEO00032
See Example 15

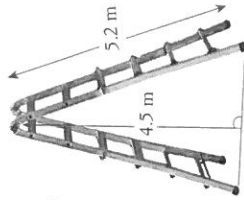
See Example 15

- 4 The slant height of a cone is 45 cm and the radius is 21 cm. Find the height of the cone correct to one decimal place.



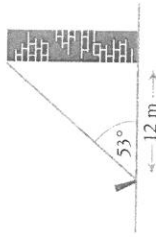
See Example 15

- 5 The arms of this fold out ladder are 5.2 m long. When fully spread, the ladder has a height of 4.5 m. How far apart are the arms of the ladder when it is fully extended?



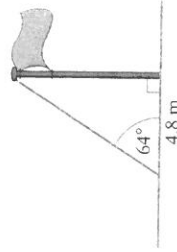
See Example 15

- 6 A survey peg is placed 12 m from the base of a wall. The base of the peg makes an angle of 53° with the top of the wall. Find the height of the wall, to the nearest centimetre.

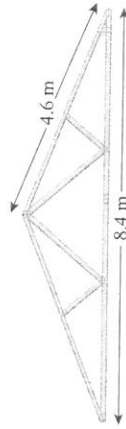


Worked solutions
Exercise 9.3
MAT09MGWS00032
See Example 16

- 7 This flagpole is held in place with a guy-wire. The wire is secured 4.8 m from the base of the pole.
a What is the height of the pole, to the nearest centimetre?
b How long is the guy-wire, to the nearest centimetre?



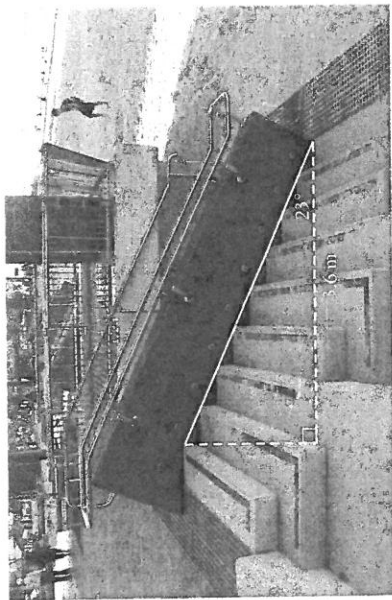
- 8 Calculate the height of this roof truss.



See Example 15

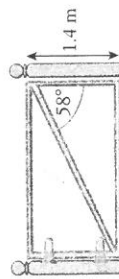
See Example 16

- 9 This set of stairs makes an angle of 23° with the horizontal. The horizontal distance between the top tread and the bottom tread is 3.6 m. What is the vertical distance between these two treads, correct to the nearest centimetre?



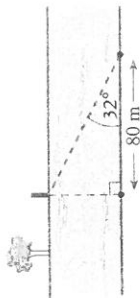
See Example 16

- 10 Find the length of this rectangular gate if the diagonal support makes an angle of 58° with the height.



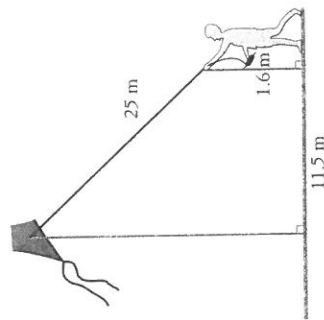
See Example 16

- 11 Two survey pegs are 80 m apart on one bank of a river. On the opposite bank is a navigation beacon. It makes an angle of 32° with one of the pegs and is directly opposite the other peg. How wide is the river (correct to 1 decimal place)?



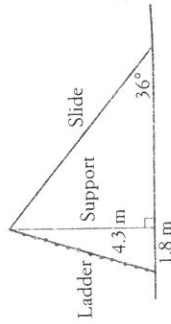
See Examples 16, 17

- 12 A kite is attached to a 25 m piece of rope, as shown. The rope is held 1.6 m above the ground and the horizontal distance between the kite and the kite flyer is 11.5 m. Find:
a the height of the kite above the ground, correct to 1 decimal place
b the angle that the string makes with the horizontal.



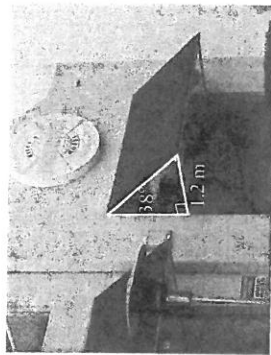
See Examples 16, 17

- 13 A playground slide makes an angle of 36° with the horizontal. The top of the slide is accessed by a ladder. The base of the ladder is 1.8 m away from a 4.3 m high vertical support as shown. Find:
a the length of the slide to the nearest centimetre
b the angle that the ladder makes with the horizontal correct to 1 decimal place.



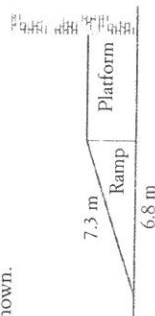
- 14 This awning is held in position by a supporting strut 1.2 m long. Calculate (correct to the nearest centimetre):

- a the distance between the top of the awning and the point where the strut meets the wall
b the length of the edge of the awning between the wall and the strut.



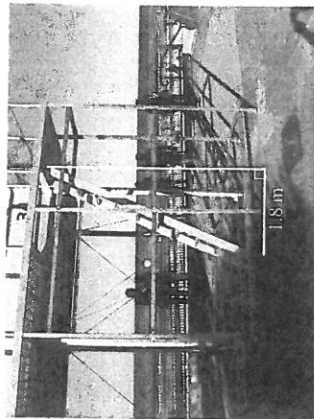
- 15 A ramp is used to deliver material to a platform as shown. Calculate:

- a the height of the ramp
b the angle that the ramp makes with the horizontal.



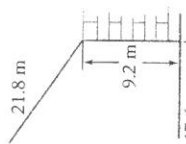
- 16 Lifesavers climb the ladder shown here to gain access to the enclosure from which they can observe the beach. The ladder is 5 m long and the base of the ladder is horizontally 1.8 m away from the top of the ladder.

- a What angle does the ladder make with the vertical?
b How high above the ground is the ladder?



- 17 A ver from pole, wall t

- 22 $UV = 166.3$ m
a the height of the wall
b the height of the pole



1 a 236.1 m	b About 82 m
2 17.7 m	3 1058.9 m
4 39.8 cm	5 5.2 m
6 10.95 m	7 a 9.84 m
b 10.95 m	8 1.9 m
9 1.53 m	10 2.2 m
11 50.0 m	12 a 23.8 m
b 62.6°	13 a 7.31 m
b 67.3°	14 a 1.54 m
b 1.95 m	15 a 2.66 m
b 21.3°	16 a 68.9°
b 4.66 m	17 a 23.9 m
b 53.8°	18 a 59.2°
b 24.4 m	19 5.4 m
20 30.4 cm	

Exercise 9.3

- 21 KL = 8.90 m

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